

Abstract—
Index Terms—

I. INTRODUCTION

Code-mixed, Roman-script social media text—commonly termed *Banglish*, in which Bangla is transliterated into Latin characters and freely interleaved with English—is ubiquitous across South Asian social platforms, yet it is poorly served by standard NLP tooling trained on formal, monolingual corpora. The core obstacle is orthographic: Banglish has no fixed spelling convention, so a single word may surface in a dozen visually and phonetically related but string-dissimilar forms, undermining tokenizers, embeddings, and classifiers that assume orthographic regularity.

We focus this problem on sentiment classification under a low-label-budget regime—the realistic deployment scenario in which thousands of unlabeled comments are readily scraped but only a small fraction can be manually annotated. We instantiate this concretely with 600 labeled examples against a pool of roughly 9,430 available training rows, a labeling rate of approximately 6.4%, reflecting the annotation budgets typical of small research teams and low-resource languages rather than the fully-labeled corpora assumed by most benchmark work.

Prior Banglish sentiment classification work has largely pursued one of two directions: data augmentation to synthetically expand labeled sets [1, 2], or fine-tuning deep transformer models on whatever labeled data is available [3]. Neither line evaluates classical two-view co-training in the original sense of Blum and Mitchell [4]—training two conditionally-independent-view classifiers that iteratively pseudo-label each other’s confident predictions—as a low-resource remedy for this specific language setting, and no prior study rigorously diagnoses why co-training helps when it does. In particular, co-training gains in the literature are typically presented as a single aggregate improvement, without decomposing how much of that gain comes from ensembling two views at all versus from the iterative pseudo-labeling process itself—a distinction this paper makes explicit.

This paper makes the following contributions:

- We reproduce and rigorously benchmark a two-view (orthographic + phonetic) co-training pipeline for Banglish sentiment classification, validating our harness against the original implementation to within 0.003 macro-F1.
- We decompose the co-training gain into an ensemble component (+0.015, from combining two views at iteration 0) and an iteration component (+0.01), showing the latter is smaller than commonly assumed.
- We show the deployed pseudo-label agreement gate’s purity advantage is not stable across iterations—it wins at iteration 0 (0.929 vs. 0.916) and loses by the final iteration (0.854 vs. 0.891)—and attribute this to declining

view independence as pseudo-labels accumulate in both pools.

- We conduct an exhaustive comparison of 21 classical classifiers across 7 ML families and find a previously untested classifier (`CalibratedClassifierCV(RidgeClassifier)`) beats the deployed choice on both views, with the advantage verified stable across label budgets (300–1,200) and under synthetic spelling noise (up to 35%).
- We identify and quantify a hard performance ceiling (~ 0.66 macro-F1 under full supervision) attributable to label-scheme noise, not model choice.

The remainder of the paper is organized as follows: Section II reviews related work; Section III describes the dataset; Section IV details the methodology; Section V reports results; Section VI discusses their implications; and Section VII concludes.

II. RELATED WORK

A. Co-training and semi-supervised learning theory

Co-training was introduced by Blum and Mitchell [4], who formalized the setting in which each labeled example can be described by two distinct feature views, and showed that if each view is individually sufficient to predict the label and the two views are conditionally independent given the label, a classifier trained on one view can generate confident pseudo-labels that usefully augment training data for a classifier trained on the other. This conditional-independence assumption is rarely satisfied exactly in practice, and Van Engelen and Hoos [5], in their survey of the broader semi-supervised learning landscape, identify the erosion of view independence over successive self-training or co-training rounds as a persistent, under-examined failure mode—one that most applied papers acknowledge in passing but few measure directly on a deployed pipeline. This is precisely the gap our paper closes: rather than reporting an aggregate co-training gain, we track the pseudo-label agreement gate’s purity iteration-by-iteration and show its advantage over the alternative literally reverses sign as the two views’ pseudo-label pools converge on each other’s outputs—an empirical instance of the theoretical concern Van Engelen and Hoos raise, observed on real Banglish data rather than in simulation.

B. Banglish and code-mixed sentiment analysis

Recent work on Bengali–English code-mixed sentiment has pursued two main strategies for coping with label scarcity and orthographic noise. Tareq et al. [1] and Sultana and Mamun [2] both rely on data augmentation—generating synthetic training examples via cross-lingual word replacement and related transformations—to expand small labeled sets and to expose the model to spelling variation. Separately, Alam et al.

[3] introduce BnSentMix, a diverse 20,000-example Bengali–English code-mixed sentiment dataset, and benchmark 14 baselines including transformer encoders further pretrained on code-mixed text, reaching 69.8% accuracy (69.1% F1) under full supervision. We position our work against this backdrop rather than in competition with it: our classical two-view pipeline, trained on only 600 labeled examples with no GPU, is not intended to surpass transformer accuracy, but to approach it at a small fraction of the label budget and zero training cost—a materially different and, for low-resource deployment settings, arguably more actionable trade-off. Where Tareq et al. and Sultana and Mamun treat orthographic noise as something to be countered with synthetically augmented spelling variants, our weakly-supervised noise model instead learns and injects noise patterns from the pipeline’s own disagreement signal, which we argue is a more targeted mechanism than augmentation applied independent of what the model actually confuses.

C. Feature engineering and classifier calibration

Our View A representation builds directly on Cavnar and Trenkle [6], who showed that character n -gram features are robust to the spelling and typographical variation that defeats word-level tokenization—the same property that motivates their use here for Banglish, where a single word can surface in many orthographic forms. Our View B phonetic encoding draws on the classical Soundex family of phonetic coding schemes [7, 8], adapted with Bangla-specific phoneme groupings. For the classifier comparison in Section V, we ground our calibration discussion in two foundational results: Platt [9], who introduced fitting a sigmoid to SVM decision scores via an internal cross-validation loop to produce probability estimates—the mechanism responsible for the substantial slowdown of `SVC(probability=True)` relative to margin-only variants—and Niculescu-Mizil and Caruana [10], who showed empirically that several standard classifiers, including SVMs and boosted trees, produce poorly calibrated probability estimates out of the box, such that an uncalibrated margin-to-probability mapping can be a materially unreliable confidence signal. These two results provide the literature basis for our finding that `CalibratedClassifierCV`-wrapped classifiers outperform their uncalibrated counterparts specifically on the confidence-threshold-gated pseudo-labeling step central to co-training, rather than leaving this as an isolated empirical observation.

Taken together, these three clusters motivate the structure of the paper: Cluster A supplies the theoretical prediction (declining view independence) that our diagnostic in Sections V and VI tests directly; Cluster B supplies the applied baseline and the augmentation-versus-learned-noise contrast that frames our contribution honestly rather than as an unqualified improvement; and Cluster C supplies the mechanistic explanation for why calibration, not classifier family alone, drives the exhaustive comparison’s result.

TABLE I
UNIFICATION OF THE THREE SOURCE LABEL SCHEMES INTO A COMMON THREE-CLASS TARGET.

Original scheme	Mapped to
BnSentMix numeric 0	Positive
BnSentMix numeric 1	Negative
BnSentMix numeric 2, 3 (“mixed”), 4	Neutral
<i>joy, happy, happiness, love</i>	Positive
<i>sadness, sad, anger, angry, disgust, fear, hate</i>	Negative
<i>surprise, surprised, neutral, mixed, none</i>	Neutral

III. DATASET

A. Source and size

We construct our working corpus by merging three publicly available resources. BnSentMix [3] contributes 20,015 rows of Bengali–English code-mixed social media text with four-way sentiment labels. The b-and-b-80K dataset [11] contributes 80,098 rows; we drop its native-Bengali-script column and retain only the Romanized (Banglish) text field, consistent with our focus on orthographic variation in Roman-script transliteration. EnBn Code-Mixed Two-Class Sentiment Balanced 100K [12] contributes 100,000 rows; the dataset carries no author attribution on its source page, and we cite it here as author-unattributed, obtained via the dataset file itself, rather than inventing a provenance. Concatenating and deduplicating the three sources by exact text match yields 186,138 unique rows. From this pool we draw a class-balanced working sample of 11,788 rows: 3,889 Negative, 4,000 Neutral, and 3,899 Positive.

B. Label unification

The three source datasets use incompatible label schemes (a five-way numeric scheme, an emotion-word scheme, and a native two-class scheme), which we unify into a single three-class Positive/Negative/Neutral target as shown in Table I.

C. Split

We hold out 20% of the balanced sample as a stratified test set (an 80/20 train/test split). From the remaining training pool we carve a stratified 600-row labeled seed set L ; the rest ($\sim 8,830$ rows) forms the unlabeled pool U used for co-training. All reported numbers are averaged across three random seeds (42, 7, 2024) to control for seed-dependent variance in the initial labeled sample.

D. Class distribution

The balanced working set is approximately uniform across the three classes by construction (Negative 3,889 / Neutral 4,000 / Positive 3,899), so per-class support in the confusion matrices reported in Section V (Figure 4) is directly comparable across classes without reweighting.

TABLE II

MAIN RESULTS: MACRO-F1 ON THE HELD-OUT TEST SET, MEAN OVER 3 SEEDS (42, 7, 2024).

Model	Macro-F1
Single-view SVM (baseline)	0.5534
Standard co-training (no WNM)	0.5767
PMVC-WNM (deployed)	0.5794
Best exhaustive candidate, View A (ridge_calibrated)	0.5652
Best exhaustive candidate, View B (ridge_calibrated)	0.5542
Full-supervision ceiling ($\sim 9,430$ labels)	0.6562

E. Representative samples

To illustrate the orthographic variation the pipeline must handle, we draw the following examples directly from the working set:

- “*ei ghrinjo opradhi dhoriye din...*” $\rightarrow$ Negative
- “*apps takay onek vlo hoise...*” $\rightarrow$ Positive

F. Preprocessing

Raw text is lowercased, URLs are stripped, and only Roman letters and digits are retained. Sequences of three or more repeated characters are collapsed to two (e.g. “sooooo” $\rightarrow$ “soo”), which normalizes the emphatic elongated typing common in social media comments without destroying word identity—collapsing to two rather than one preserves the doubled-letter spellings that are themselves legitimate in Bangla transliteration. Exact-text-match deduplication is applied across the merged pool before sampling, preventing a single virally-copied comment from inflating the apparent size of its class.

G. Known limitations and biases

Two limitations of this dataset construction are worth stating explicitly rather than glossing over. First, the label unification folds two semantically distinct phenomena—genuinely mixed or ambivalent sentiment, and surprise—into a single Neutral class; this is a modeling simplification, and we present the full-supervision performance ceiling of 0.656 macro-F1 (Table II) as direct evidence that this label scheme, not model capacity alone, bounds what any classifier can achieve on this task. Second, the three source datasets differ in genre—YouTube and Facebook comments, e-commerce product reviews, and general social media text respectively—and merging them may introduce domain-shift noise between the labeled seed set and the test distribution, depending on which sources happen to dominate each stratified split.

IV. METHODOLOGY

V. RESULTS

VI. DISCUSSION

VII. CONCLUSION

REFERENCES

- [1] M. Tareq, M. F. Islam, S. Deb, S. Rahman, and A. A. Mahmud, “Data-augmentation for Bangla-English code-mixed sentiment analysis: Enhancing cross linguistic

TABLE III

VIEW A / VIEW B CLASSIFIER COMPARISON (MEAN OVER 3 SEEDS). LOWER BRIER SCORE INDICATES BETTER-CALIBRATED CONFIDENCE ESTIMATES.

View	Classifier	Macro-F1	Brier	Fit (s)
A	svm_platt	0.5624	0.1807	1.435
A	logreg	0.5613	0.1802	0.729
A	svm_calibrated	0.5596	0.1813	0.120
A	svm_softmax	0.5534	0.1960	0.018
B	logreg	0.5522	0.1845	1.314
B	svm_softmax	0.5488	0.2002	0.007
B	complement_nb	0.5340	0.1952	0.005
B	random_forest	0.5245	0.1907	1.492

TABLE IV

FEATURE N-GRAM RANGE COMPARISON AT THE DEPLOYED 5,000-FEATURE BUDGET (MEAN OVER 3 SEEDS).

View	Config	Macro-F1
A	char(2,3)	0.5611
A	char(3,4) [deployed]	0.5534
B	word(1,1)	0.5575
B	word(1,2) [deployed]	0.5522

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- [8] —, “Index,” U.S. Patent 1,435,663, Nov. 1922.
- [9] J. Platt, “Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods,” *Advances in Large Margin Classifiers*, vol. 10, no. 3, pp. 61–74, 1999.
- [10] A. Niculescu-Mizil and R. Caruana, “Predicting good probabilities with supervised learning,” in *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, 2005, pp. 625–632.

TABLE V
PSEUDO-LABEL GATE COMPARISON, CONFIDENCE THRESHOLD FIXED AT 0.55 (MEAN $\pm$ STD OVER 3 SEEDS).

Gate	Macro-F1	Pseudo-labels	Model Purity	Family	Holdout F1	CV F1	Fit (s)
agreement_and_confident (pmvc_wnm)	0.5794 $\pm$ 0.0094	7191	0.859	ridge_calibrated	linear	0.5652 $\pm$ 0.0064	0.6688
view_a_only (standard)	0.5767 $\pm$ 0.0101	5783	0.881	logreg	linear	0.5613 $\pm$ 0.0089	0.6599
agreement_or_one_very_confident	0.5739 $\pm$ 0.0058	21827	0.757	bagging_linsvc	tree_ensemble	0.5601 $\pm$ 0.0013	0.6631
agreement_no_threshold	0.5701 $\pm$ 0.0077	35100	0.709	linsvc_calibrated	linear	0.5596 $\pm$ 0.0065	0.6642
or_union	0.5701 $\pm$ 0.0077	35100	0.673	knncv15	instance_based	0.5541 $\pm$ 0.0069	0.6199

TABLE VI
CUMULATIVE SINGLE-VIEW (VIEW A) IMPROVEMENT LADDER, EACH ROW ADDS ONE CHANGE ON TOP OF THE PREVIOUS (MEAN OVER 3 SEEDS).

Configuration	Macro-F1
0. baseline (svm_softmax, char(3,4), 5k feat)	0.5534 $\pm$ 0.0120
1. max_features=50000	0.5729 $\pm$ 0.0036
2. + min_df=2	0.5733 $\pm$ 0.0016
3. + class_weight=balanced	0.5741 $\pm$ 0.0024
4. + C tuned (C=0.3)	0.5751 $\pm$ 0.0028
5. + word(1,2) UNION char(3,5) [50k total]	0.5830 $\pm$ 0.0019
6. + CalibratedClassifierCV(LinearSVC)	0.5849 $\pm$ 0.0032

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- [12] Author unattributed, “EnBn code-mixed two-class sentiment balanced 100K,” Dataset file; no author attribution given on the source page, 2026.

TABLE VII
TOP 5 OF 21 CANDIDATES TESTED, VIEW A (MEAN OVER 3 SEEDS).

Model	Family	Holdout F1	CV F1	Fit (s)
ridge_calibrated	linear	0.5542 $\pm$ 0.0076	0.6446	0.06
logreg	linear	0.5522 $\pm$ 0.0076	0.6381	0.58
linsvc_calibrated	linear	0.5519 $\pm$ 0.0074	0.6420	0.05
linsvc_softmax	linear	0.5488 $\pm$ 0.0065	0.6359	0.01
bagging_linsvc	tree_ensemble	0.5471 $\pm$ 0.0065	0.6377	0.05

TABLE IX
LABEL-BUDGET SENSITIVITY: TEAM’S DEPLOYED MODEL VS. BEST-FOUND MODEL (MEAN OVER 3 SEEDS).

Labeled seed	Team A	Best A	Team B	Best B
300	0.5376	0.5466	0.5176	0.5126
600	0.5534	0.5652	0.5522	0.5542
900	0.5669	0.5813	0.5701	0.5729
1200	0.5908	0.6017	0.5830	0.5848

TABLE X
SPELLING-NOISE ROBUSTNESS: MACRO-F1 UNDER INJECTED PHONETIC-VARIANT CORRUPTION (MEAN OVER 3 SEEDS).

Noise rate	Team A	Best A	Team B	Best B
0%	0.5534	0.5652	0.5522	0.5542
20%	0.5486	0.5614	0.5516	0.5535
35%	0.5444	0.5554	0.5524	0.5548

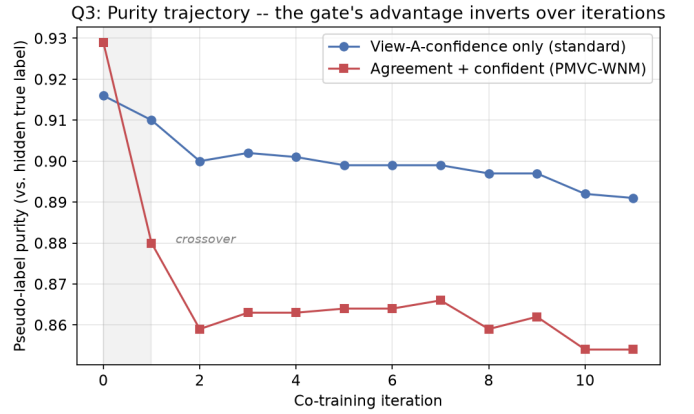


Fig. 1. Pseudo-label purity by co-training iteration, seed 42. The agreement gate’s purity advantage appears only at iteration 0 and inverts by the final iteration.

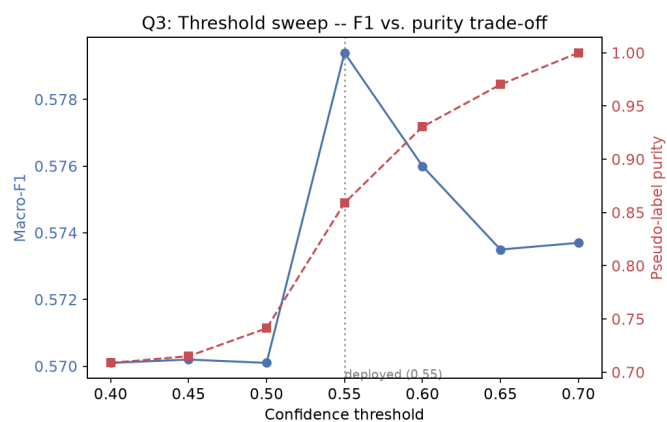


Fig. 2. Macro-F1 (left axis) and pseudo-label purity (right axis) vs. confidence threshold on the deployed agreement gate.

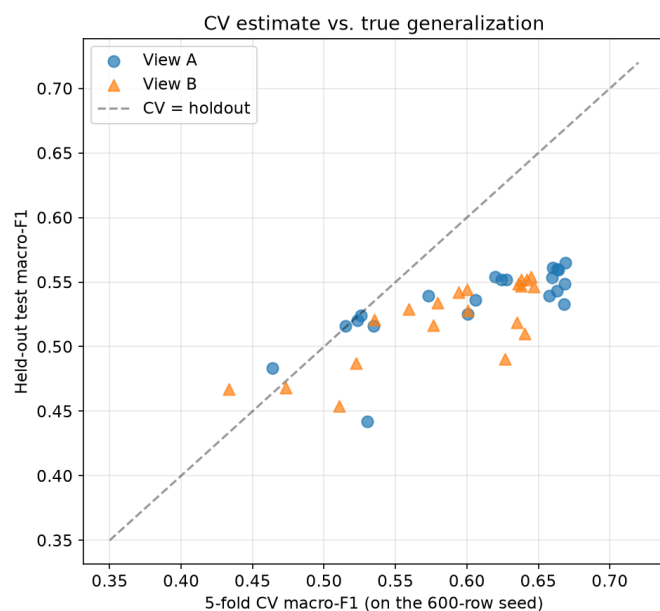


Fig. 3. Every candidate's 5-fold CV estimate vs. its held-out test score. Points below the diagonal generalize worse than their CV estimate suggested.

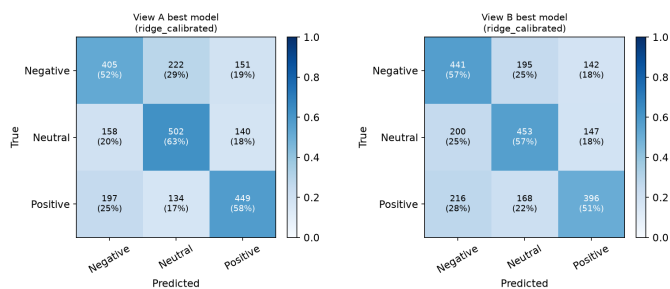


Fig. 4. Row-normalized confusion matrices, best model per view, seed 42.